AI Solution Architecture Canvas

Your capstone, built a little each day. Fill one section at the end of each day; by Friday it is your design. One short line per box is enough.

Section	Guiding questions	Your answer (one short line per box)	Taught on
1 Problem, users and KPIs	Who is the user? Which decision or task improves? What is the baseline today? 2 to 3 KPIs (business and technical). What does 'good' look like?		Day 1, 2
2 Data and sources	What data exists (structured, unstructured, streaming)? Quality, freshness, labels, owners. Sensitive data and who may see what?		Day 2, 3
3 Approach: rules, ML, RAG, agent or hybrid	Which of the seven gates apply? Why this rung of the ladder and not a simpler one? Fine-tune, prompt or RAG, and why?		Day 1, 2, 3
4 Retrieval and answer quality	Chunking, embeddings, vector or hybrid search, filters and access control, reranking. How do you measure quality (golden set, Recall@k)? Hallucination controls and refusals.		Day 3
5 Agents, tools and MCP	Does it need an agent? Which tools (search, SQL, APIs)? Which pattern (ReAct, plan-and-execute, router, human-in-the-loop)? Step limits, approvals, MCP servers.		Day 4
6 MLOps, monitoring and release	Versioning and tracking, CI/CD, quality gates, monitoring (data, model, system, business), drift and retrain triggers, A/B or canary release.		Day 4

7 Cloud services and deployment	Which cloud services (models, search, ML platform, agents)? Serverless, containers or managed APIs? Region and data residency. Integration with CRM, ERP, data warehouse.		Day 4, 5
8 Responsible AI, security and compliance	Bias and fairness checks, explainability, prompt-injection and leakage defences, PII handling, GDPR / RBI / HIPAA relevance, governance and human oversight.		Day 5
9 Cost, latency, risks and next steps	Rough cost per 1,000 requests, latency target (p95), top 3 risks with mitigations, what you would build first (MVP) and what you would defer.		Day 2, 5

Team: _____ Scenario: _____ Date: _____